CHAPTER 4

Polynomial equations

Quadratics in $\mathbb{C}$, factorisation, number of roots

Objectives. Solve in $\mathbb{C}$ a quadratic equation with real coefficients; factorise a polynomial by $z - a$ when a is a root; solve a cubic equation when one root is known; know that a polynomial of degree n has at most n roots.

Prerequisites. Quadratics in $\mathbb{R}$ (completed square form, discriminant, sum and product of the roots); chapters 1 and 2; sum of the terms of a geometric sequence.

Fluency

Fluency · No calculator, five minutes.

1. Compute the discriminant of $2x^2 - 3x + 5$.
2. Factorise $x^2 - 5x + 6$.
3. Expand $(x - 2)\left(x^2 + 2x + 4\right)$.
4. Compute $P(2)$ for $P(x) = x^3 - 3x^2 + 4$.
5. Write $1 + q + q^2 + \cdots + q^n$ as a quotient, for $q \neq 1$.
6. Give the sum and the product of the roots of $x^2 - 7x + 10$.

Introductory activity

Activity 1 · One root hides two others 35 min . Consider $P(z) = z^3 - 2z^2 + 4z - 8$.

1. Plot on a calculator the curve of $x \mapsto x^3 - 2x^2 + 4x - 8$ on $[-3, 4]$. How many real solutions does the equation $P(x) = 0$ seem to have? Locate one.
2. Check that $P(2) = 0$.
3. Find real numbers a, b, c such that, for every z, $P(z) = (z - 2)\left(az^2 + bz + c\right)$. Expand the right-hand side and identify the coefficients.
4. Deduce all the complex solutions of $P(z) = 0$.
5. Compute the sum and the product of the three solutions. Compare with the coefficients of P.
6. How many complex solutions can a polynomial of degree 3 have at most? State a conjecture for degree n.

Observing is not proving: the properties conjectured here are established in the course.

Course notes

Quadratics with real coefficients

Property · Solving in $\mathbb{C}$. Let a, b, c be three real numbers with $a \neq 0$, and $\Delta = b^2 - 4ac$. The equation $az^2 + bz + c = 0$ has:

1. if $\Delta > 0$, two real solutions $\dfrac{-b - \sqrt{\Delta}}{2a}$ and $\dfrac{-b + \sqrt{\Delta}}{2a}$;
2. if $\Delta = 0$, one real solution $-\dfrac{b}{2a}$;
3. if $\Delta < 0$, two conjugate complex solutions $z_1 = \dfrac{-b - i\sqrt{-\Delta}}{2a}$ and $z_2 = \dfrac{-b + i\sqrt{-\Delta}}{2a}$.

In every case, $az^2 + bz + c = a(z - z_1)(z - z_2)$, $z_1 + z_2 = -\dfrac{b}{a}$ and $z_1 z_2 = \dfrac{c}{a}$ (with $z_1 = z_2$ if $\Delta = 0$).

Proof. The completed square form $az^2 + bz + c = a\left[\left(z + \frac{b}{2a}\right)^2 - \frac{\Delta}{4a^2}\right]$ still holds in $\mathbb{C}$. If $\Delta < 0$, write $\frac{\Delta}{4a^2} = \left(\frac{i\sqrt{-\Delta}}{2a}\right)^2$, then factorise the difference of two squares. A product of complex numbers is zero if and only if one of the factors is zero (since every nonzero complex number is invertible). Sum and product are obtained by expanding $a(z - z_1)(z - z_2)$ and identifying. □

Example 1 · Solving $z^2 - 2z + 5 = 0$. $\Delta = 4 - 20 = -16 = (4i)^2$, so $z = \frac{2 \pm 4i}{2}$: the solutions are $1 - 2i$ and $1 + 2i$.

Factorisation by $z - a$

Definition · Polynomial function, root. A *polynomial function* of degree n is a function $P : z \mapsto a_n z^n + \cdots + a_1 z + a_0$ defined on $\mathbb{C}$, where $a_0, \ldots, a_n$ are complex numbers and $a_n \neq 0$. A complex number α such that $P(\alpha) = 0$ is a *root* of P.

Key point · Property taken for granted. If a polynomial function is zero for every complex number z, then all its coefficients are zero. Consequently, two equal polynomial functions have the same coefficients: we may *identify* the coefficients.

Property · Factorisation of $z^n - a^n$. For all complex numbers z, a and every integer $n \geq 1$:

$$z^n - a^n = (z - a)\left(z^{n-1} + az^{n-2} + \cdots + a^{n-2}z + a^{n-1}\right) = (z - a)\sum_{k=0}^{n-1} a^k z^{n-1-k}.$$

Proof. Expand: $(z - a)\sum_{k=0}^{n-1} a^k z^{n-1-k} = \sum_{k=0}^{n-1} a^k z^{n-k} - \sum_{k=0}^{n-1} a^{k+1} z^{n-1-k}$. Setting $j = k + 1$ in the second sum, it becomes $\sum_{j=1}^{n} a^j z^{n-j}$. The terms with index 1 to $n - 1$ cancel; what remains is the term $k = 0$ of the first sum, z^n, and the term $j = n$ of the second, a^n. We obtain $z^n - a^n$. □

Property · Factorisation by $z-\alpha$. Let P be a polynomial function of degree $n \geq 1$ and α a root of P. There is a polynomial function Q of degree $n-1$ such that, for every complex number z, $P(z) = (z-\alpha)Q(z)$.

Proof. Write $P(z) = \sum_{k=0}^{n} a_k z^k$. Since $P(\alpha) = 0$,

$$P(z) = P(z) - P(\alpha) = \sum_{k=1}^{n} a_k \left(z^k - \alpha^k\right).$$

By the previous theorem, each $z^k - \alpha^k$ can be written $(z-\alpha)Q_k(z)$ with Q_k a polynomial of degree $k-1$ with leading coefficient 1. So $P(z) = (z-\alpha)Q(z)$ with $Q = \sum_{k=1}^{n} a_k Q_k$, of degree $n-1$ with leading coefficient $a_n \neq 0$. □

Property · Number of roots. A polynomial function of degree $n \geq 1$ has at most n roots.

Proof. By induction on n. A polynomial function of degree 1, $a_1 z + a_0$ with $a_1 \neq 0$, has exactly one root, $-\frac{a_0}{a_1}$. Assume the property holds in degree n, and let P have degree $n+1$. If P has no root, we are done. Otherwise, let α be a root: $P(z) = (z-\alpha)Q(z)$ with Q of degree n. If $P(\beta) = 0$, then $\beta = \alpha$ or $Q(\beta) = 0$. Since Q has at most n roots, P has at most $n+1$. □

Property · Conjugate roots. If P has real coefficients and α is a root of P, then $\overline{\alpha}$ is also a root of P.

Proof. Since the coefficients are real, $P\left(\overline{\alpha}\right) = \sum a_k \overline{\alpha}^k = \overline{\sum a_k \alpha^k} = \overline{P(\alpha)} = 0$. □

Method · Cubic equation with a known root. Factorise by $z-\alpha$ by identifying the coefficients of $Q(z) = az^2 + bz + c$, then solve $Q(z) = 0$. Example: $z^3 + 8 = 0$ has the root -2 and $z^3 + 8 = (z+2)\left(z^2 - 2z + 4\right)$. The quadratic has discriminant $-12 = \left(2i\sqrt{3}\right)^2$, hence the solutions -2, $1 - i\sqrt{3}$, $1 + i\sqrt{3}$.

Watch out. The discriminant of a quadratic with real coefficients no longer tells whether the equation has solutions in $\mathbb{C}$: it always has. It tells whether they are real or conjugate complex numbers.

Exercises

Exercise 4.1 · Quadratics. Solve in $\mathbb{C}$: $z^2 + 9 = 0$; $z^2 - 4z + 13 = 0$; $2z^2 + 2z + 1 = 0$; $z^4 - 1 = 0$.

Exercise 4.2 · Reducing to a quadratic.

1. Solve in $\mathbb{C}$ the equation $z^4 + 4z^2 - 5 = 0$ by setting $Z = z^2$.
2. Solve in $\mathbb{C}$ the equation $z + \dfrac{1}{z} = 1$, for $z \neq 0$.

Exercise 4.3 · An obvious root. Let $P(z) = z^3 - 3z^2 + 4z - 2$.

1. Check that 1 is a root of P.
2. Determine the polynomial Q such that $P(z) = (z - 1)Q(z)$.
3. Solve $P(z) = 0$ in $\mathbb{C}$.

Exercise 4.4 · A non-real root. Let $P(z) = z^3 + z^2 + 4z + 4$.

1. Check that $2i$ is a root of P. Deduce another root without computation.
2. Expand $(z - 2i)(z + 2i)$, then factorise $P(z)$ and solve $P(z) = 0$.

Exercise 4.5 · Factorising $z^3 - 27$.

1. Factorise $z^3 - 27$ by $z - 3$ using the course.
2. Solve in $\mathbb{C}$ the equation $z^3 = 27$ and plot the solutions in the complex plane. What do you notice?

Exercise 4.6 · A parameter. Let m be a real number. Consider the equation (E_m): $z^2 - 2z + m = 0$.

1. Depending on the value of m, give the number and the nature of the solutions of (E_m).
2. Assume $m > 1$. Let A and B be the images of the solutions. Determine m so that the triangle OAB is equilateral.

Exercise 4.7 · Recovering a polynomial. Determine the polynomial P of degree 3, with real coefficients and leading coefficient 1, such that $1 + i$ is a root of P and $P(0) = 4$.

Problem

Problem · Coefficients and roots of a cubic polynomial. Let $P(z) = z^3 + az^2 + bz + c$ be a polynomial with complex coefficients, with three roots z_1, z_2, z_3 (not necessarily distinct), so that $P(z) = (z - z_1)(z - z_2)(z - z_3)$ for every complex number z.

1. By expanding, prove that
$$z_1 + z_2 + z_3 = -a, \qquad z_1z_2 + z_1z_3 + z_2z_3 = b, \qquad z_1z_2z_3 = -c.$$
2. Check these relations on the polynomial of the activity, $z^3 - 2z^2 + 4z - 8$.
3. Prove that $z_1^2 + z_2^2 + z_3^2 = a^2 - 2b$.
4. Consider $R(z) = z^3 - 2z^2 + 3z - 5$, which we take for granted can be written $(z - z_1)(z - z_2)(z - z_3)$. Compute $z_1^2 + z_2^2 + z_3^2$ without solving the equation.
5. Deduce that the roots of R are not all real.

Beyond the syllabus · The d'Alembert-Gauss theorem. Every polynomial of degree $n \geq 1$ with complex coefficients factorises in $\mathbb{C}$ as a product of n factors of degree 1. This theorem is not part of the syllabus: it is admitted in question 4 and is not assessed.